

Question: Write down the properties and applications of χ^2 dist.

Answer:

Properties of χ^2 distribution: The properties of χ^2 distribution are given below —

i) The pdf of χ^2 -distribution with n d.f is

$$f(x^2) = \frac{e^{-x^2/2} (x^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma(\frac{n}{2})} ; 0 \leq x^2 < \infty$$

ii) The range of χ^2 -distribution is zero to infinity.

iii) The mean of χ^2 -distribution is equal to its d.f and the variance is equal to its twice d.f.

iv) This distribution is positively skewed and leptokurtic.

v) Its mode is $n-2$.

vi) χ^2 distribution approaches normal distribution as $n \rightarrow \infty$.

vii) If $X \sim N(0,1)$ then $X^2 \sim \chi^2$ with 1 d.f.

Applications of χ^2 distribution: χ^2 -distribution has a large no. of applications in statistics, some of which are enumerated below—

i) To test if the hypothetical value of the population variance is $\sigma^2 = \sigma_0^2$ (say).

ii) To test the 'goodness of fit'.

iii) To test the independence of attributes.

iv) To test the homogeneity of independent estimates of the population variance.

v) To test the homogeneity of independent estimates of the population correlation coefficient.

d.f = degrees of Freedom

Question: Write down the properties and uses of t-distribution.

Answer:

Some properties of t-distribution are given below—

i) The density function of t-distribution with n d.f is

$$f(t) = \frac{1}{\sqrt{n} \beta(\frac{n}{2}, \frac{1}{2}) (1 + \frac{t^2}{n})^{\frac{n+1}{2}}} ; -\infty < t < \infty$$

ii) It is an exactly sampling distribution derived from standard normal and χ^2 -distribution.

iii) All odd ordered moments are equal to zero.

iv) For large d.f, t-distribution tends to standard normal distribution.

v) If $n=1$, t-distribution follows Cauchy distribution.

Some uses of t-distribution are given below—

i) It is used to test the ^{population} sample mean ($\bar{x}$) differs significantly from the hypothetical value of the population mean.

ii) It is used to test the significance of the difference between two ^{population} sample means.

iii) It is used to test the significance of an observed sample correlation co-efficient and sample regression co-efficient.

iv) It is used to test the significance of observed partial and multiple correlation co-efficients.

In the limit as $n_2 \rightarrow \infty$, we have

$$\frac{\sqrt{\frac{n_1+n_2}{2}}}{n_2^{n_1/2} \sqrt{\frac{n_2}{2}}} \rightarrow \frac{(n_2/2)^{n_1/2}}{n_2^{n_1/2}} = \frac{1}{2^{n_1/2}} \quad \left| \because \frac{\sqrt{n+k}}{\sqrt{n}} \rightarrow n^k \text{ as } n \rightarrow \infty \right.$$

$$\text{Also } \lim_{n_2 \rightarrow \infty} \left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1+n_2}{2}} = \lim_{n_2 \rightarrow \infty} \left[\left(1 + \frac{n_1}{n_2} F\right)^{n_2}\right]^{1/2} \cdot \lim_{n_2 \rightarrow \infty} \left(1 + \frac{n_1}{n_2} F\right)^{n_1/2}$$

$$= e^{n_1 F/2} \cdot 1 = e^{x^2/2} \quad \left| \because x^2 = n_1 F \right.$$

Hence in the limit, the pdf of $x^2 = n_1 F$ becomes

$$f(x^2) = \frac{\left(\frac{n_1}{2}\right)^{\frac{n_1}{2}} \frac{1}{2^{n_1/2}} e^{-x^2/2}}{\sqrt{\frac{n_1}{2}}} \left(\frac{x^2}{n_1}\right)^{\frac{n_1}{2}-1} \cdot \frac{1}{n_1} \quad \left| \because J = \left|\frac{dF}{dx^2}\right| = \frac{1}{n_1} \right.$$

$$\Rightarrow |J| = \frac{1}{n_1}$$

$$= \frac{e^{-x^2/2} (x^2)^{\frac{n_1}{2}-1}}{2^{n_1/2} \sqrt{\frac{n_1}{2}}} \quad ; 0 \leq x^2 < \infty$$

which is the pdf of x^2 -distribution with n_1 d.f.

Question: Write down the properties and uses of F-distribution.

Answer: The properties of F-distribution are given below—

i) The density function of F-distribution with n_1, n_2 d.f is

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1+n_2}{2}}} \quad ; 0 \leq F < \infty$$

ii) The mean, variance and mode of this distribution are

$$\mu_1' = \frac{n_2}{n_2-2}, \quad n_2 > 2; \quad \mu_2' = \frac{n_2^2}{(n_2-2)^2} \left[\frac{2(n_1+n_2-2)}{n_1(n_2-4)} \right], \quad n_2 > 4$$

and $F = \frac{(n_1-2)n_2}{n_1(n_2+2)}$ respectively.

iii) If $X \sim F(n_1, n_2)$ then $Y = \frac{1}{X} \sim F(n_2, n_1)$.

iv) The t^2 -dist is the special case of the F-dist where the numerator has only one d.f.

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v) If denominator d.f n_2 alone tends to infinity then $n_1 F$ tends to χ^2 distribution with n_1 d.f.

v) As the d.f increases for both numerator and denominator the F-distribution tends to normal distribution.

Some uses are given below —

i) To test the homogeneity of means in analysis of variance.

ii) To test the hypothesis on equality of two population variance

$$H_0: \sigma_1^2 = \sigma_2^2.$$

iii) To test the significance of correlation ratio.

iv) To test the significance of multiple correlation coefficient.

v) To test the non-linearity of regression.

vi) To test the significance of inter class correlation coefficient.